

PWM Control of a Magnetic Suspension System

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Abstract—Magnetic levitation of a metallic sphere motivates and inspires students to investigate and understand the fundamental principles of electrical engineering, such as magnetic design, circuit design, and control theory. This paper describes the operation of a pulsewidth modulation converter in a magnetic suspension system. The pulsewidth-modulated (PWM) converter illustrates modern principles of power electronics, such as PWM control, current-mode control, averaged and linearized models of switched-mode converters, and power supply design. The reduced size of the heat sink in the switched-mode converter compared with its linear amplifier counterpart clearly illustrates improved efficiency. The experimental system described levitates a 6-cm-diameter, 0.8-kg metallic sphere.

Index Terms—Control engineering education, electromagnetics, magnetic levitation, pulsewidth modulation (PWM), switched-mode power supplies.

NOMENCLATURE

a	Decay constant for coil inductance.
d	Duty cycle.
f	Switching frequency.
G	Gain of differential amplifier.
G_S	Gain of light sensor (V/m).
g	Acceleration due to gravity 9.81 m/s^2 .
i	Coil current.
K_p	Proportional gain.
L	Coil inductance at equilibrium.
$L(x)$	Inductance at position x .
M	Mass of sphere.
N	Number of coil turns.
R	Coil resistance.
R_I	Gain of current sensor (V/A).
T	Switching period ($1/f$).
T_d	Derivative time.
v_c	Control voltage.
V_p	Peak value of the PWM ramp.
v_{PD}	Output voltage of PD controller.
v_x	Output voltage of light sensor.
V_s	Supply voltage.
x	Distance between coil and magnet.
Φ	Diameter of suspended sphere.
μ_o	Permeability of free space $4\pi \times 10^{-7} \text{ H/m}$.
$\omega_n = \sqrt{g/a}$	Natural frequency (r/s).

Notation: The instantaneous variable is lower case; the quiescent or average value is upper case; and the incremental component is lower case with a prime, e.g., $d = D + d'$. The Laplace

transform of the incremental variable is upper case with the variable s , e.g., $D(s)$.

I. INTRODUCTION

THE stable suspension of a metallic sphere by an electromagnet has been employed as an educational tool for many decades. It serves to visually demonstrate and reinforce the underlying principles of electromagnetism, control theory, and circuit design [1]–[3]. More recently, fuzzy logic has been applied to the control of the system [4], [5]. The system is a powerful illustration of associated applications, such as magnetically levitated trains and frictionless bearings.

The material described in this paper applies mainly to electrical engineering courses in circuit theory, circuit design, control theory, power electronics, and sensors (current and position). Therefore, it spans the entire curriculum from freshman level through graduate studies.

The magnetic suspension system is an unstable nonlinear system. Reference [1] describes the linearization of the plant (the coil/sphere) by examining perturbations around the operating point. Compensation is achieved by implementing proportional plus derivative (PD) control. The student has the opportunity to design the control system using classical tools such as root locus and Bode plots. The direct implementation of the control strategy in a practical system provides a strong motivation to the student to study further compensation techniques.

Traditionally, the control of the magnetic suspension system has been achieved with a linear amplifier. This paper proposes the use of a switching converter with pulsewidth modulation (PWM). Switched-mode power supplies (SMPSs) have largely replaced linear power supplies because of improved efficiencies. Typically, the heat sink in the system described here is less than one tenth by volume of its linear counterpart. Switches are nonlinear devices. Averaged models are needed to study the steady-state behavior of switching converters. Moreover, linearized models are needed to study dynamic behavior, for control-loop analysis and for compensation in an unstable system [6]. The paper illustrates all of these principles in the design of a switching converter and introduces the student to switching circuits, PWM control, and current feedback.

A number of experiments are described to validate the theoretical principles. The student is expected to design the PD controller; the step response of the system may be predicted (using MATLAB) and then measured. The measurement of coil inductance as a function of ball position is described. The position sensor is characterized, and typical PWM waveforms are also measured.

Manuscript received March 12, 2001; revised January 30, 2003.

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Digital Object Identifier 10.1109/TE.2004.827831

The authors' experience indicates that the pedagogical impact is immediate and that the students are attracted to study many diverse principles through experimentation with the magnetic suspension system described.

II. THE ELECTROMAGNETIC SYSTEM

The principal components of the magnetic suspension system are shown in Fig. 1. The position of the ball is sensed by the optical system, which is employed in a feedback loop to control the position of the ball by means of a PWM converter.

The characteristics of the magnet are described in detail in [1] and are summarized here for completeness. The inductance of the coil with a ball at a distance x is given by

$$L(x) = L_1 + L_0 e^{-x/a}. \quad (1)$$

L_1 is the value of inductance when the ball is removed, and a is a constant based on the coil dimensions. $L_0 + L_1$ is the inductance when the ball is in contact with the coil. For optimum coil design, the diameter of the coil core is 0.8Φ , where Φ is the diameter of the ball and $a = \Phi/9$. For $\Phi = 6.0$ cm, the following values apply with $N = 1000$ turns [1]:

$$\begin{aligned} L_0 &= 0.229 \text{ H} \\ L_1 &= 0.349 \text{ H} \\ a &= 6.66 \text{ mm.} \end{aligned}$$

The magnetic co-energy of the system is a function of the coil current i and separation x so that

$$W' = \frac{1}{2} L(x) i^2. \quad (2)$$

The force of magnetic origin acting on the ball is

$$f(i, x) = \frac{\partial W'}{\partial x} = -\frac{L_0}{2a} i^2 e^{-x/a}. \quad (3)$$

In static equilibrium, the gravitational force on the ball Mg , where M is the mass of the ball and g is acceleration because of gravity, balances this force. Equating these forces at $x = X$, $i = I$ yields the coil current in the equilibrium position

$$I = \sqrt{2Mg \frac{a}{L_0 e^{-X/a}}}. \quad (4)$$

The 6-cm ball has $M = 0.8$ kg, and $I = 1.69$ A, and $X = 1$ cm. A 1.12-mm-diameter wire will maintain the steady-state temperature rise below 30°C , resulting in a coil resistance of 4.3Ω . The inductance of the coil at equilibrium is $L = 0.4$ H.

III. THE DYNAMIC SYSTEM

The overall block diagram is shown in Fig. 2. The set point determines the steady-state position of the ball. The individual blocks are based on the transfer functions of each part of the system, which are derived from small-signal analysis as described in this section and in Section IV. The position is sensed optically, and a voltage is generated by a photodiode. The control signal to the PWM controller is generated by the amplified

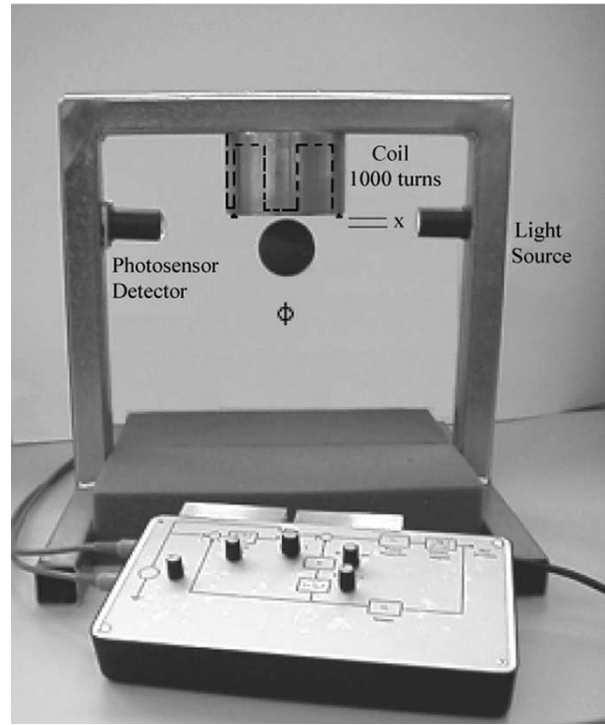


Fig. 1. The electromagnetic suspension system.

output of the PD controller and a signal that is proportional to the coil current. The control signal determines the duty cycle of the switch, which in turn controls the voltage to the coil. The main purpose of the inner control loop is to improve the dynamic response of this highly inductive coil circuit. This section describes the block diagram for the plant, the position sensor, the current sensor, and the PD controller.

A. Coil Current/Position

The plant transfer function for perturbations in coil current and separation is [1]

$$\frac{X(s)}{I(s)} = \frac{-2g}{s^2 - \omega_n^2} \quad (5)$$

$$\omega_n = \sqrt{\frac{g}{a}}. \quad (6)$$

The transfer function is completely determined by the equilibrium current and the constant a . From Section II, $I = 1.69$ A, $a = 6.66$ mm.

B. Position Sensor

The position of the ball is sensed optically. The current in the photodiode is a function of the level of light detected, depending on the position of the ball. The output of the photodiode is amplified, and the resulting output voltage varies monotonically with the ball position. Perturbations in the ball position about the equilibrium point are linearly related to the variation in output voltage, measured and shown in Section V-C. The circuit details are shown in Fig. 3. Section IV establishes the relationship between the control voltage, the output of the PWM converter, and the coil current.

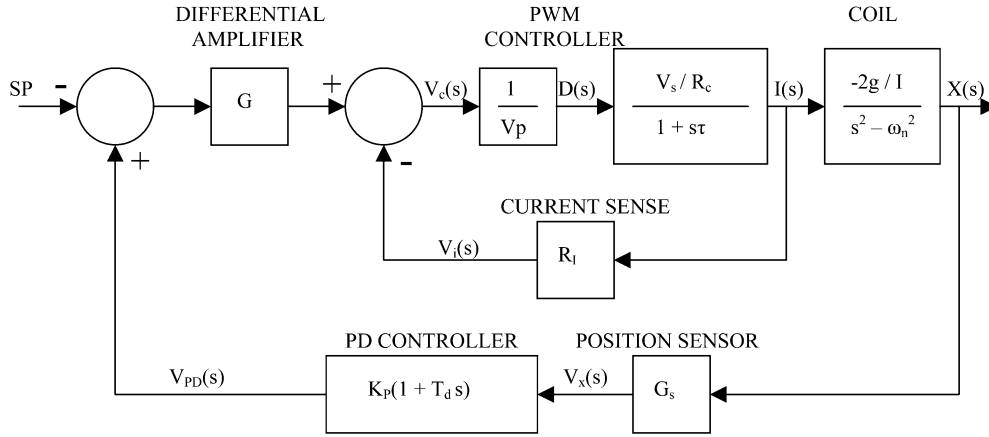


Fig. 2. Block diagram of the compensated system.

C. The Current Sensor

The current sensor consists of a current mirror and differential amplifier as shown in Fig. 3. The action of the current mirror, combined with the output of the differential amplifier, yields

$$v_i = \frac{R_2}{R_1} \cdot \frac{R_c}{R_e} R_s \cdot i = R_I i. \quad (7)$$

D. The PD Controller

The output of the light sensor detector circuit is fed into the PD controller as shown in Fig. 3. By inspection

$$\begin{aligned} v_{PD} &= \frac{R_p}{R_3} (1 + R_d C_d s) v_x \\ &= K_p (1 + T_d s) v_x. \end{aligned} \quad (8)$$

The output from the current sense circuit is subtracted from the output of the PD circuit in a differential amplifier to generate the control voltage (v_c). The control voltage is compared to the ramp voltage to generate the gate signal, which controls the switch Q as illustrated in Fig. 3.

IV. PWM CONVERTER

In this section, the transfer functions of the PWM converter are derived based on the established techniques for power electronic circuits. One assumes that the student would not be familiar with averaging and linearizing modeling techniques; therefore, these are described in more detail.

A. Steady-State Operation

The basic circuit, in Fig. 4(a), consists of the coil (inductance L and resistance R), the metal–oxide–semiconductor field-effect transistor (MOSFET) switch Q , and the diode D , which protects the MOSFET during turn-off. In the absence of the diode, the rapid decrease in inductor current would result in excessively high inductor and transistor voltages.

The components can be rearranged as shown in Fig. 4(b). This is the basic buck converter configuration, except that the

output capacitor is missing. Vorperian [6] describes the process of averaging and linearizing the canonical switching cell.

There are two modes of operation: when the switch is closed and when the switch is open. The inductor voltage and current waveforms are shown in Fig. 5.

Consider steady-state operation. The average voltage across the inductor is zero, so that the average output across R is

$$V_0 = DV_s. \quad (9)$$

The steady-state or average output voltage is a function of the duty cycle D of the switch.

B. Averaged Circuit Model

The steady-state analysis described previously is based on a cycle-by-cycle analysis of the circuit. Averaging the conditions in the canonical switching cell may adequately represent the transient behavior of the circuit. The canonical switching cell consists of the switch Q and the diode D . Since the averaged circuit is linear (the nonlinear switches have been removed), linear circuit theory and control theory may be employed to analyze the behavior of the circuit. The L/R time constant is approximately 100 ms which is much greater than the switching period (20 kHz), which is 0.05 ms. Rather than simulating the circuit on a cycle-by-cycle basis, an averaged model may be used, provided $L/R \gg T$. The averaged circuit greatly reduces the computation time for transient analysis.

The canonical switching cell and its equivalent circuits are shown in Fig. 6. The switch Q operates with a duty cycle D , the output voltage $v_{cp} = Dv_{ap}$, and the input current $i_a = Di_c$. A “transformer” with a turns ratio $1:D$ may represent the canonical switching cell, as shown in Fig. 6. The circuit, which implements the average switch condition, is shown in Fig. 6(b).

The averaged circuit for the magnetic suspension system of Fig. 4 is shown in Fig. 7. By inspection, the average output (coil) voltage is DV_s . The set point is established by the supply voltage V_s and the steady-state duty cycle D . The steady-state current in the equilibrium position is DV_s/R . Typically, $V_s = 24$ V, $R = 4.3$ Ω , and $D = 0.3$ for a set-point current of 1.69 A.

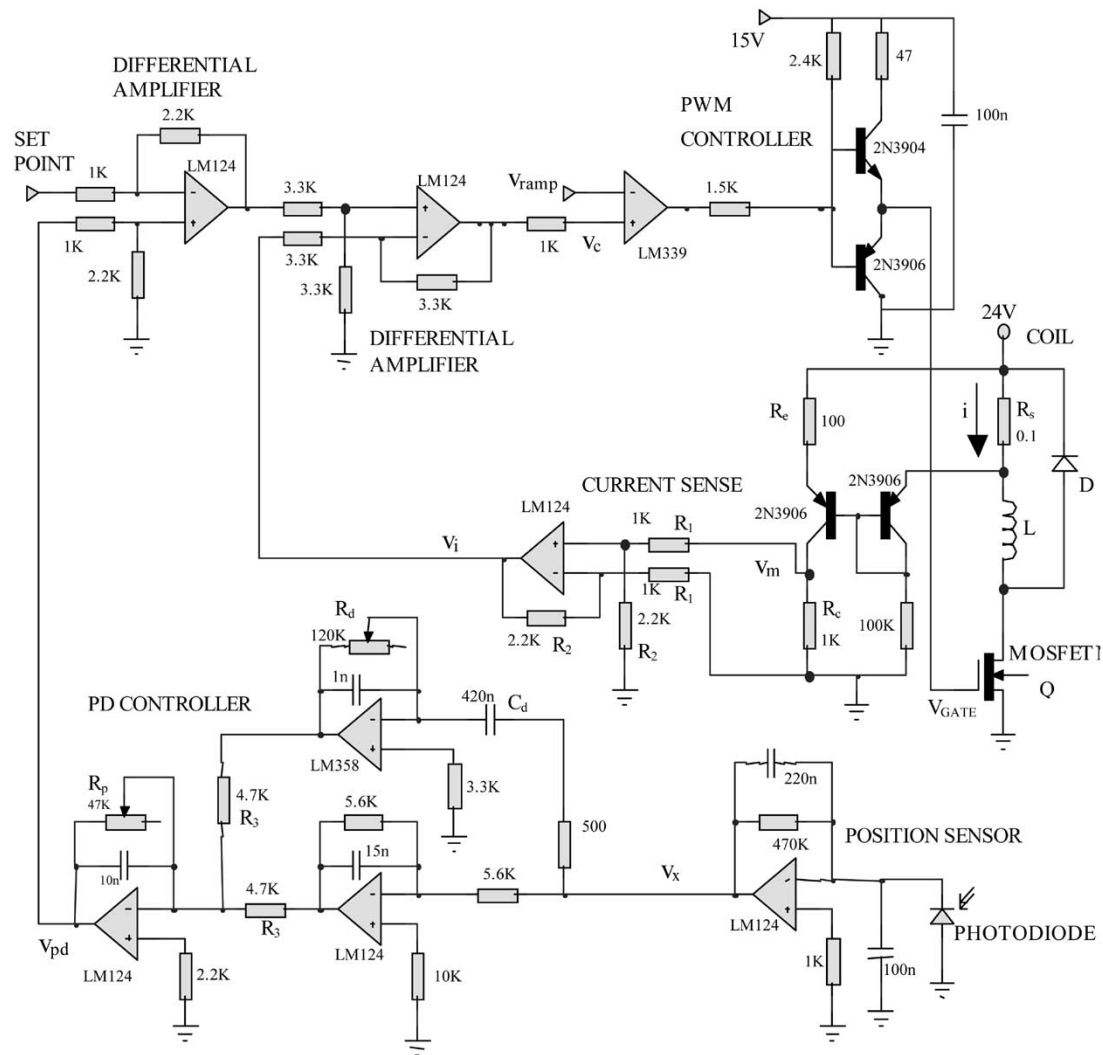


Fig. 3. Circuit for magnetic suspension system.

C. Small-Signal Analysis/Linearized Circuit

The dynamic behavior of the circuit is based on perturbations about the steady-state or average values of voltage, current, and duty cycle, as follows:

$$\begin{aligned} v_0 &= V_0 + v'_0 \\ d &= D + d' \\ i_L &= I + i'_L \\ v_s &= V_s + v'_s \end{aligned}$$

Referring to the canonical switching cell of Fig. 6, the steady-state and small-signal components may be separated, as follows:

$$\begin{aligned} v_{cp} &= dv_s = (D + d')(V_s + v'_s) \\ &= DV_s + Dv'_s + d'V_s + d'v'_s. \end{aligned} \quad (10)$$

Neglecting the product of small quantities such as $d'v'_s$

$$v_{cp} = DV_s + Dv'_s + d'V_s \quad (11)$$

and

$$i_a = di_c = DI_c + Di'_c + d'I_c. \quad (12)$$

The linearized circuit is obtained by replacing all quantities by their perturbations from the nominal or steady-state, as shown in Fig. 8.

The transfer function, relating coil current to duty cycle, is obtained by setting $v'_s = 0$, and taking Laplace transforms

$$\frac{I(s)}{D(s)} = \frac{\frac{V_S}{R}}{1 + s\tau}; \quad \tau = \frac{L}{R}. \quad (13)$$

Equation (13) is the transfer function relating the coil current to the duty cycle of the PWM controller. The relationship between the duty cycle and the control voltage is derived next.

D. The PWM Controller

Evidently, by varying the duty cycle of the switch, the coil current can be controlled. In pulsewidth modulation, the control voltage (v_c) is compared with a repetitive ramp waveform, as

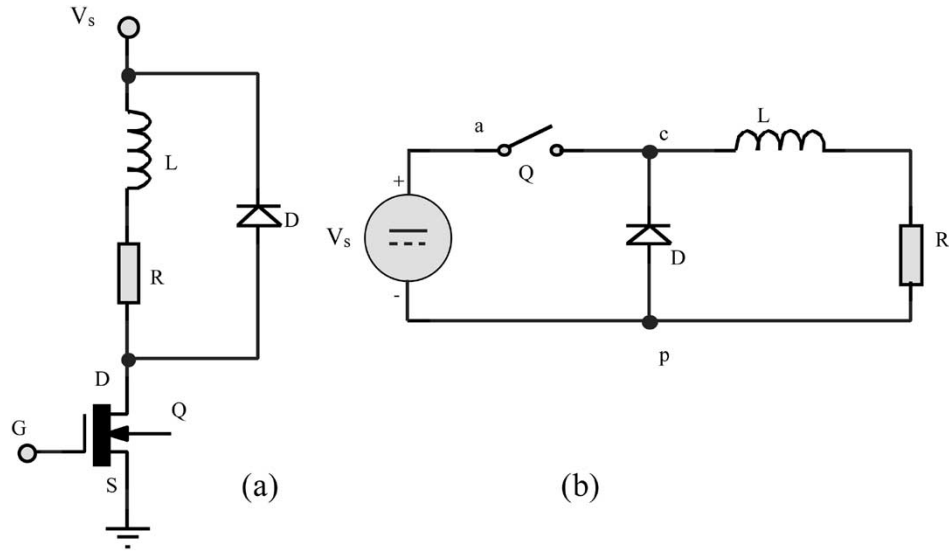


Fig. 4. (a) Coil/switch arrangement and (b) rearranged circuit.

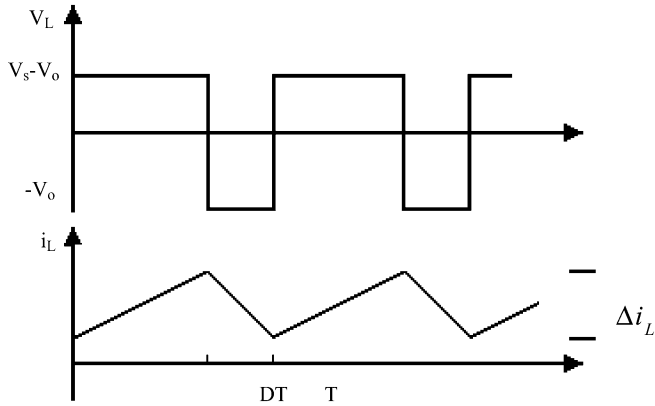


Fig. 5. Circuit waveforms, inductor voltage, and current.

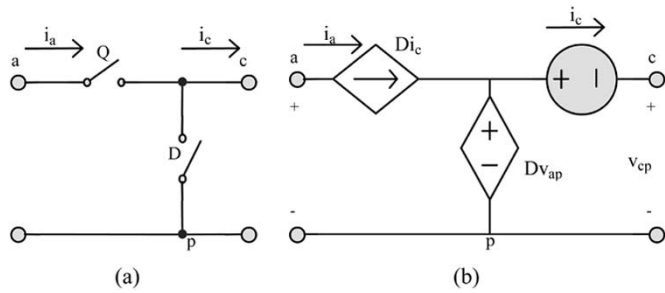


Fig. 6. The canonical switching cell: (a) switch configuration and (b) averaged circuit.

shown in Fig. 9. The output of the comparator is applied to the switch Q . By inspection, the duty cycle is

$$d = \frac{v_c}{V_p} \quad (14)$$

and the transfer function is

$$\frac{D(s)}{V_c(s)} = \frac{1}{V_p} \quad (15)$$

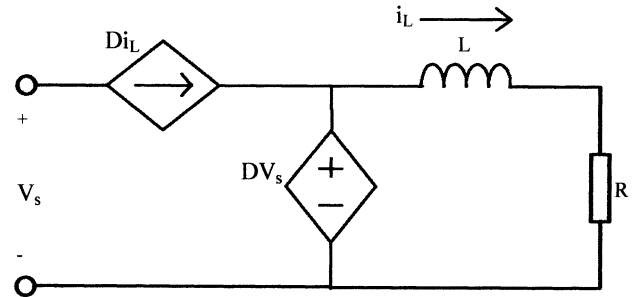


Fig. 7. Averaged circuit for the magnetic suspension system.

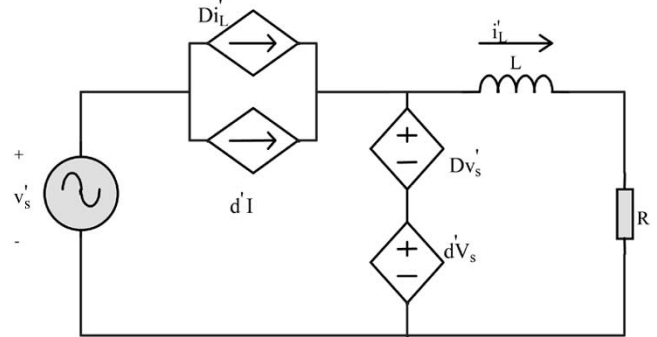


Fig. 8. Linearized (small-signal) equivalent circuit.

where V_p is the peak value of the ramp waveform. The circuit implementation of the ramp generator is described in the Appendix.

V. STUDENT EXPERIMENTS

The analysis of Sections II–IV leads to a number of experiments, both theoretical and practical, which allow students to investigate and understand the principles involved. The exercises consist of paper studies using established tools such as MATLAB followed by experimental measurements. In this way, the theoretical principles are reinforced by practical validation.

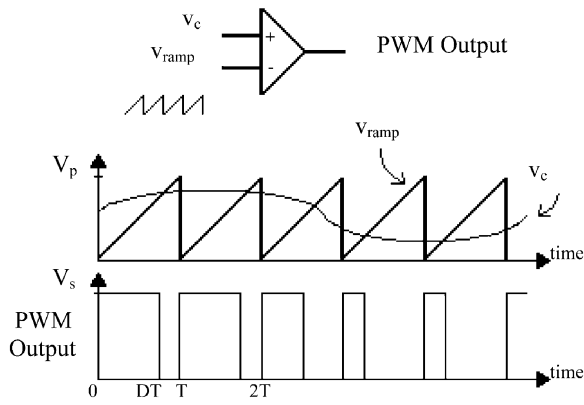


Fig. 9. PWM control, circuit, and waveforms.

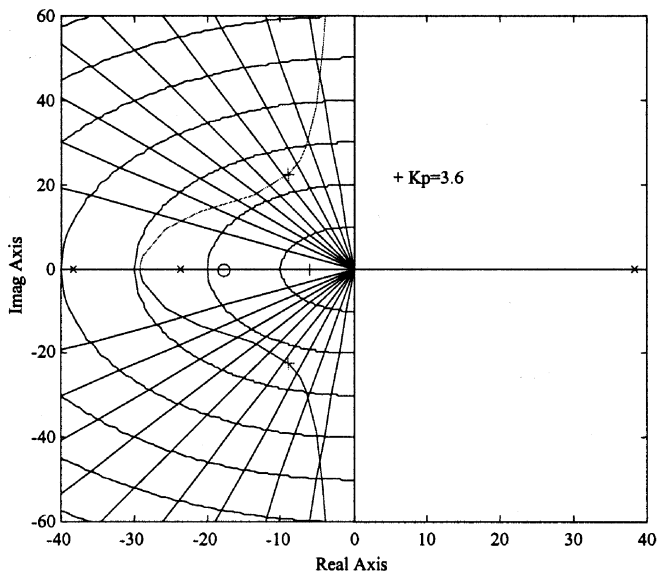


Fig. 10. Root locus of the experimental system, damping ratios in increments of 0.1, loci of natural frequencies in increments of 10 r/s.

The plant transfer function is used in many textbooks as an example of an unstable system; the student can see the results of a design based on classical methods such as root locus.

The magnetic suspension system was constructed with the following parameters:

- Mass of sphere $M = 0.8$ kg;
- Diameter of sphere $\Phi = 6.0$ cm;
- Equilibrium gap $X = 1$ cm;
- Coil inductance $L = 0.4$ H;
- Coil resistance $R = 4.3$ Ω ;
- Natural frequency $\omega_n = 38.38$ $\frac{r}{s}$;
- Optical sensor gain $G_s = 69$ $\frac{V}{m}$;
- PD Controller $K_p = 3.6$;
- $T_d = 0.056$ s;
- Current sensor gain $R_I = 2.2$ $\frac{V}{A}$;
- PWM gain $\left(\frac{1}{V_p}\right) G_{PWM} = 0.1$ V^{-1} ;
- Equilibrium current $I = 1.69$ A;
- Supply voltage $V_s = 24$ V;
- Ramp voltage $V_p = 10$ V;
- Differential amplifier gain $G = 2.2$.

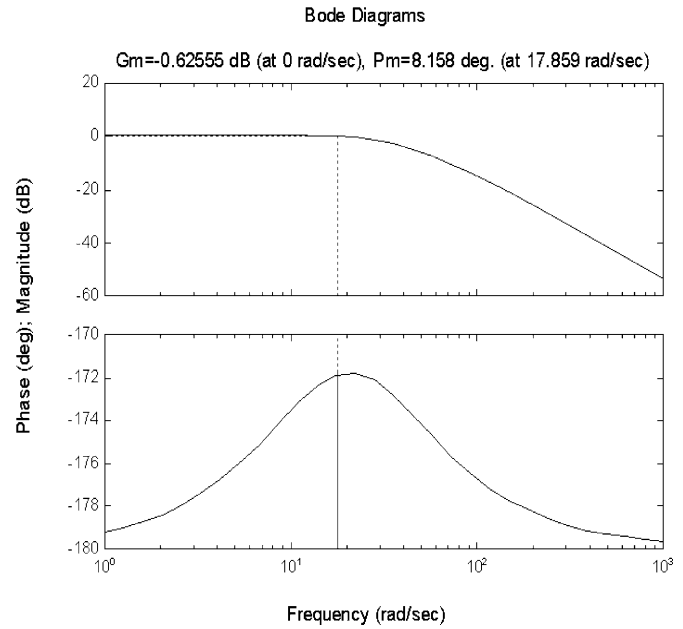


Fig. 11. Bode plot of the compensated system.

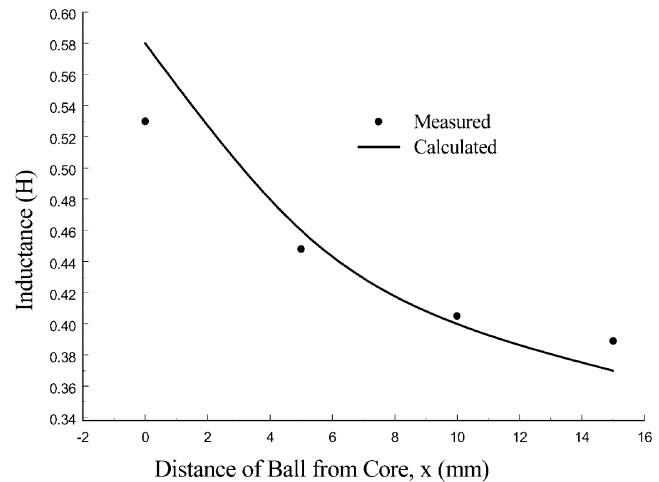


Fig. 12. Coil inductance as a function of separation.

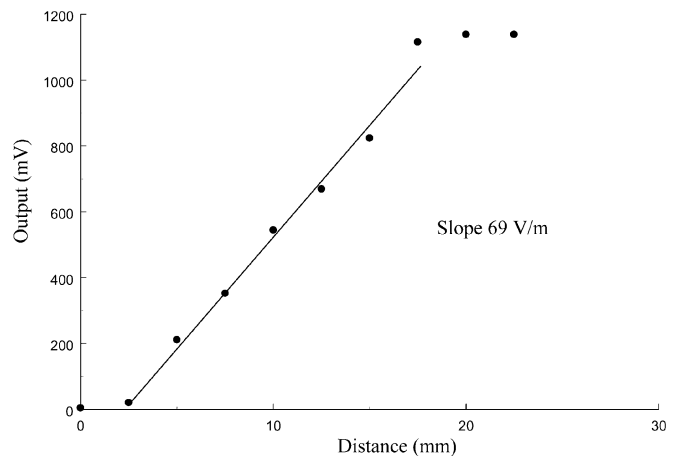


Fig. 13. Characteristic of position sensor.

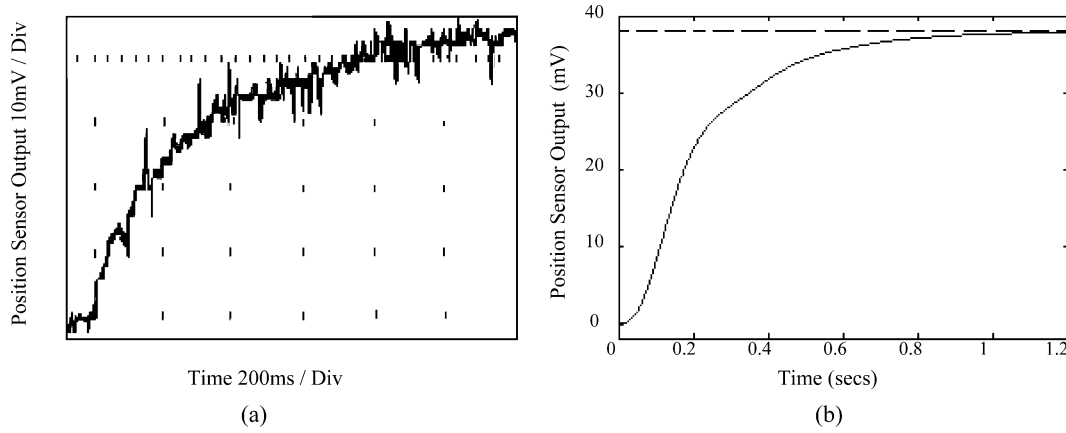


Fig. 14. Step response: (a) measured, scale 10 mV/div, 200 ms/div and (b) predicted by MATLAB.

A. Root Locus and Bode Plot

The root locus, associated with the block diagram with PD control, is shown in Fig. 10. The Bode plot, in Fig. 11, reveals that the proportional gain $K_p = 3.6$ yields a phase margin of 8° at the unity gain crossover frequency of 17.9 r/s. As an exercise, the student could devise an improved compensation system to increase the phase margin.

B. Coil Inductance Measurement

The inductance of the coil as a function of x can be measured by applying a step of voltage to the coil, observing the exponential response of the current, and measuring the time constant $L(x)/R$ of the response. The resistance of the coil was measured as 4.3Ω . With $x = \infty$, $L_1 = 0.372$ H; and with $x = 0$, $L_0 + L_1 = 0.534$ H; therefore, $L_0 = 0.162$ H. At $x = 1$ cm, $L(x) = 0.405$ H. The decay constant a is found to be 6.3 mm from (1). These numbers compare very well with the theoretical values in Section II. The graph of $L(x)$ versus x is shown in Fig. 12 for calculated and measured values.

C. Sensor Gain

The characterization of the position sensor is investigated by measuring the output of the position sensor for different values of ball position x . The characteristic is shown in Fig. 13. The gain is obtained from the slope of the characteristic.

$$G_s = \frac{v'_s}{x'} \quad (16)$$

G_s is 69 V/m as measured in Fig. 13.

D. Step Response

A step is applied, in the form of a square wave of 0.5 Hz, at the set point, which has been adjusted so that the ball is in the middle of the operating range in order to achieve maximum response. The response is observed at the output of the position sensor and is compared with the predicted response in Fig. 14(b), which was generated in MATLAB for a 10-mV input.

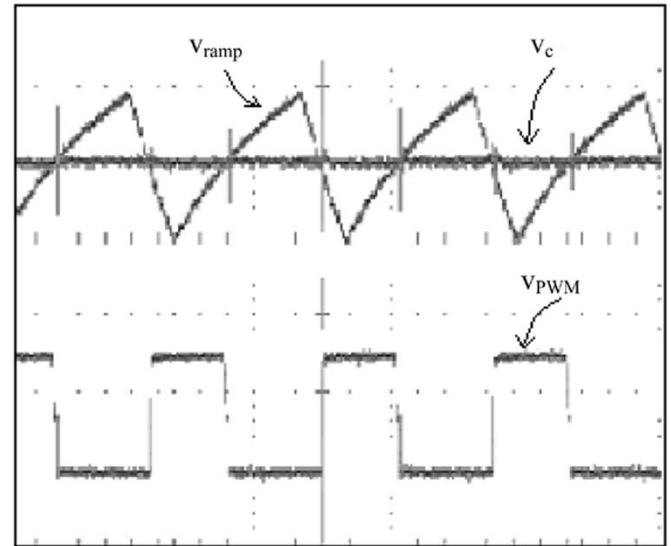


Fig. 15. PWM waveforms: control voltage, ramp, gate signal scale 5 V/div, and 20 μ s/div.

E. PWM Waveforms

The ramp waveform, the control voltage, and the gate pulse in the experimental system are captured on the oscilloscope and shown in Fig. 15. These can be compared with the expected waveforms in Fig. 9. The switching sequence which shows a duty cycle of 0.4 can be observed. This duty cycle is somewhat higher than the predicted value; however, unlike experiments B and C, where the ball can be physically held in place, this experiment must be carried out with the ball in suspension; and it is difficult to set the separation exactly.

VI. CONCLUSION

A magnetic suspension system with a PWM converter has been described. The transfer function of each component in the system has been derived from first principles. The magnetic design, the dynamic response, the converter, and the PWM controller have been described in detail. The compensation of the control system has been accomplished by employing pole placement techniques. Current mode control and lead compensation,

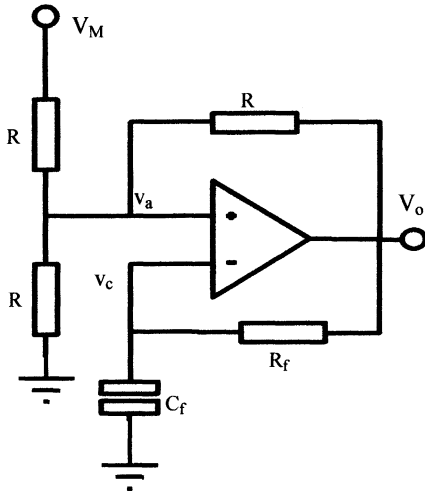


Fig. 16. Schmitt trigger.

in the form of a PD controller, have been described. Several student experiments have been described which reinforce the theory.

The student can design his or her own compensation and test the results on the system. One of the salient features of the system is that the entire control loop must be properly designed to obtain stable and robust operation. (Interested readers can contact the author for further details (wghurley@ieee.org).)

APPENDIX THE RAMP GENERATOR

The combination of an integrator circuit with a Schmitt trigger circuit may be used to generate the ramp for the PWM circuit. The Schmitt trigger can be realized by applying positive feedback around an operational amplifier, as shown in Fig. 16. The operational amplifier has a large positive voltage at the noninverting input terminal. The amplifier is saturated with a positive output voltage V_M .

The voltage v_a is by superposition

$$v_a = \frac{V_M}{3} + \frac{V_o}{3} = \frac{2}{3}V_M, \quad V_o = +V_M$$

$$= 0, \quad V_o = -V_M. \quad (A1)$$

In order to force the circuit to change state, the input at the inverting terminal must rise to the value at v_a (currently at $2/3 V_M$), and the circuit enters its linear operating region. The resulting positive feedback forces the output to $-V_M$ and v_a to 0. The inverting terminal then approaches 0 to switch the output back to $+V_M$, thus completing the oscillating cycle; the waveforms are shown in Fig. 17.

The charging of the capacitor may be approximated to a linear rise if the time constant $R_f C_f \ll T$

$$v_c(t) = \frac{V_M}{R_f C_f} t, \quad 0 < t < \frac{T}{2}$$

$$= V_M - \frac{V_M}{R_f C_f} t, \quad \frac{T}{2} < t < T \quad (A2)$$

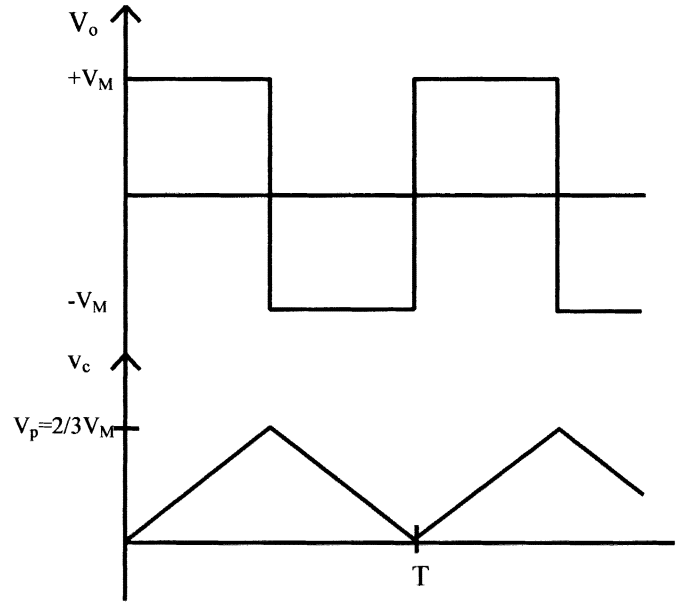


Fig. 17. Ramp waveforms.

and the switching period is

$$T = \frac{4}{3} R_f C_f$$

for $V_M = 15$ V, $V_p = 2/3 V_M = 10$ V.

ACKNOWLEDGMENT

The following undergraduate students have made significant contributions to this work: S. Carter, National University of Ireland, Galway; S. Arnold, Enserb, France; M. Uls and K. Kroger, Fachhochschule Konstanz, Germany; J. Thomas and J. Aldrete, Massachusetts Institute of Technology; and G. Debray, Université de Nantes, France. The authors would like to thank Dr. Lim of Ngee Ann Polytechnic, Singapore, for his support. The support of Enterprise Ireland is gratefully appreciated.

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